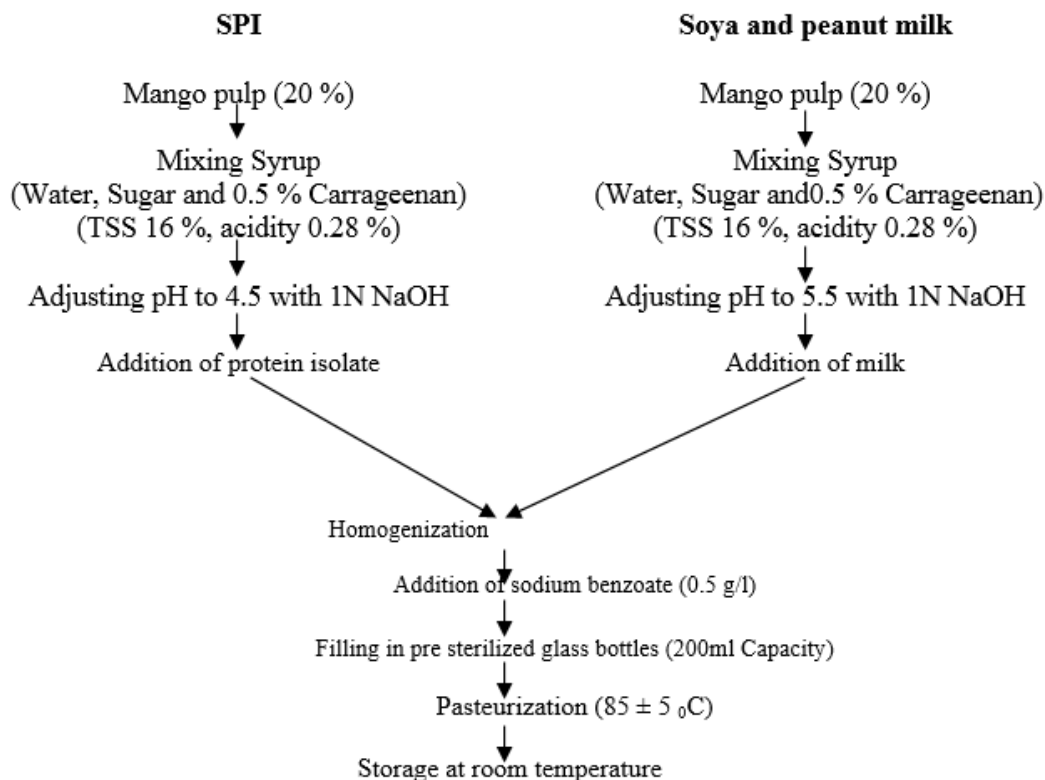




Flow sheet for preparation of plant protein fortified mango RTS drink

RTS beverages were analyzed for changes in chemical constituents and sensory quality at monthly interval for three months. Total soluble solids were estimated at ambient temperature by Abbe's refractometer (0-95%) or by hand refractometer (0-32%, Erma, Japan) and the values were expressed as per cent TSS. Acidity and ascorbic acid were analyzed by the methods of AOAC, 2005. Crude protein was estimated using Micro-Kjeldhal method with KELPLUS nitrogen estimation system. Non-enzymatic browning was recorded for stored product, by the procedure as described by Ranganna (1995) [14]. pH of the product was estimated by pH meter (Model: CL 54 digital Toshniwal Instruments Mfg. Pvt. Ltd., India). The sample (in case of pulp and squash) was diluted 1:10 for pH determination.

Results and discussion

a. PH and TSS

There was a slight decrease in pH of RTS with increasing storage period. There was no significant difference in pH of the variants of RTS containing sucrose or fructose (Tables 1). Fruit juices have a low pH because they are comparatively rich in organic acids (Tasnim *et al.*, 2010) [20]. The pH is negative function of natural acidity, thus decrease in pH is accompanied with increase in acidity of fruit juice during storage (Rehman *et al.*, 2014) [16]. There was a slight but significant increase in TSS with increasing storage period of RTS. There was no significant effect observed in TSS of variants of RTS where sucrose was replaced by fructose (Tables 1). The increase in TSS might be due to hydrolysis of insoluble polysaccharides into simple and soluble sugars (Woodroof and Luh, 1975; Majumdar, 2011) [21, 10]

Table 1: Effect of different treatments and storage period on pH and TSS (%) of protein fortified mango RTS

Treatments	Storage period (days)									
	pH					TSS (%)				
	0	30	60	90	Mean	0	30	60	90	Mean
T ₀	3.9	3.9	3.7	3.7	3.8	16.0	16.2	16.3	16.4	16.2
T ₁	3.9	3.9	3.7	3.8	3.8	16.0	16.2	16.4	16.5	16.3
T ₂	4.5	4.4	4.3	4.1	4.3	16.0	16.3	16.5	16.5	16.3
T ₃	4.5	4.4	4.3	4.2	4.4	16.0	16.4	16.6	16.6	16.4
T ₄	5.5	5.5	5.4	5.4	5.5	16.0	16.1	16.3	16.7	16.3
T ₅	5.5	5.5	5.3	5.2	5.4	16.0	16.3	16.6	16.6	16.4
T ₆	5.5	5.5	5.4	5.2	5.4	16.0	16.2	16.4	16.7	16.3
T ₇	5.5	5.5	5.3	5.3	5.4	16.0	16.1	16.4	16.7	16.3
Mean	4.9	4.8	4.7	4.6		16.0	16.2	16.4	16.6	

CD at 5 % Treatment = 0.15.; Storage = 0.12; Treatment × Storage = 0.28 Treatment = NS; Storage = 0.17; Treatment × Storage = NS

T₀ (Control with Sucrose); T₁ (Control with Fructose); T₂= T₀+ SPI; T₃= T₁ + SPI; T₄=T₀ + PM; T₅= T₁+PM; T₆= T₀+SM; T₇= T₁+SM; SPI= Soya Protein isolate; PM= Peanut milk; SM= Soya milk